Week 5: Sheet

A. De7a El Gamed

2 seconds, 256 megabytes

We all know how competitive our friends $De7a$ and $Essawi$ can get. They have been competing to decide who is the faster 3x3 speedcuber.

Now $Zula$ wants this to end right now and realized there is no better way than to give them both a problem and see who can solve it first and he asked you to try to beat them both to the punch.

Given Q queries, each one consisting of two integers L and R, you need to count the number of divisors of the number M, where M is the product of all numbers from L to R $(M = \Pi_{i=L}^{R} i)$. As the answer maybe very large print it modulo $10^9 + 7$.

The clock is ticking as $De7a$ and $Essawi$ are already started coding their solutions.

Input

The first line of input consists of a single integer $Q (1 \leq Q \leq 100)$ the number of queries Zula will ask.

Each of the next Q lines contains two integers L and R $(1 \leq L \leq R \leq 10^5)$.

Output

For each test case output the answer to the query as Zula described modulo $10^9 + 7$.

input

```
1
2 3
```

output

```
4
```

input

```
2
7 7
3 5
```

output

```
2
12
```

In the **first** test case the product is the number 6 which has 4 divisors 1, 2, 3 and 6.

How modulo is used:

A few distributive properties of modulo are as follows:

1. $(a + b)\%c$ = $((a\%c) + (b\%c))\%c$
2. $(a \times b)\%c$ = $((a\%c) \times (b\%c))\%c$
3. $(a - b)\%c$ = $((a\%c) - (b\%c))\%c$

B. Minimum Cost

1 second, 256 megabytes

Lolo was playing with his n cubes, each cube has a number written on it, He wanted to pick m cubes so that the sum of all the numbers written on the picked cubes is minimized, The steps are :

1. He picks a cube.

2. He adds the number written on it.

3. He multiplies the cube number by two.

4. He puts it back.

He repeats the steps m times, Find the minimum sum of the m cubes.

Input

The first line contains $n, m (1 \leq n \leq 100), (1 \leq m \leq 20)$ the number of cubes and the number of cubes he wants to pick respectively.

The second line contains the numbers written on each cube $v (1 \leq v \leq 10^9)$.

Output

Print the minimum sum of the picked cubes.

input

```
4 3
1 2 3 4
```

output

```
5
```

C. A Highway To...

2 seconds, 256 megabytes

Given a road of length N and an initial speed K (for the sake of the simplicity of the problem, speed limit K means the minimum number of hours needed to pass 1 kilometer of the road is K).

If a car is going to drive through the road, it starts with an initial speed K from point 0, If it comes across a new speed limit sign, it follows the new speed limit on that sign until the driver meets a new sign or reaches the end of the road.

Initially, the road doesn't have any speed limit signs, You will be given Q queries of 2 types with the following format:

- $1\ A\ B$, Insert a new sign on the road at point A with speed limit B.
- $2\ A$, Remove the sign located at point A on the road.

After each query, print the minimum number of hours needed to pass the road if the driver follows the speed limits.
For examples, take a look at the notes.

Input

The first line contains two integers N and K $(3 \leq N \leq 10^9, 1 \leq K \leq 10^9)$. the length of the road and the initial speed limit.

The second line contains one integer Q $(1 \leq Q \leq 2 \times 10^5)$. The number of queries.

Then Q lines follow. The i-th line contains the i-th query in the same format as in the problem statement. It is guaranteed that queries are always valid (for query $type$ 1, given the new position A $(1 < A < N)$ and the new speed limit B $(1 \leq B \leq 10^9)$ it is guaranteed that the new position doesn't have any signs, and for query $type$ 2, given the position A $(1 < A < N)$ it is guaranteed that there is a sign at position A).

$(1 \leq type \leq 2)$.

Output

Print Q lines. In the i-th line after performing the i-th query, print the minimum number of hours the driver will need to pass the road if he follows the speed limit.

input

```
10 1
5
1 5 2
2 5
1 3 5
1 8 10
2 3
```

output

```
15
10
38
48
28
```

input

```
3 1
1
1 2 1000000000
```

output

```
1000000002
```

In the first example the answer to :

1. The first query is : $1 \times 5 + 2 \times 5 = 15$.
2. The second query is : $1 \times 10 = 10$.
3. The third query is : $1 \times 3 + 7 \times 5 = 38$.
4. The fourth query is : $1 \times 3 + 5 \times 5 + 2 \times 10 = 48$.
5. The fifth query is : $1 \times 8 + 2 \times 10 = 28$.

D. Prepend and Append

1 second, 256 megabytes

Timur initially had a binary string[†] s (possibly of length 0). He performed the following operation several (possibly zero) times:

- Add 0 to one end of the string and 1 to the other end of the string. For example, starting from the string 1011, you can obtain either 010111 or 110110.

You are given Timur's final string. What is the length of the **shortest** possible string he could have started with?

[†] A binary string is a string (possibly the empty string) whose characters are either 0 or 1.

Input

The first line of the input contains an integer t ($1 \leq t \leq 100$) — the number of testcases.

The first line of each test case contains an integer n ($1 \leq n \leq 2000$) — the length of Timur's final string.

The second line of each test case contains a string s of length n consisting of characters 0 or 1, denoting the final string.

Output

For each test case, output a single nonnegative integer — the shortest possible length of Timur's original string. Note that Timur's original string could have been empty, in which case you should output 0.

input

```
9
3
100
4
0111
5
10101
6
101010
7
1010110
1
1
2
10
2
11
10
1011011010
```

output

```
1
2
5
0
3
1
0
2
4
```

In the first test case, the shortest possible string Timur started with is 0, and he performed the following operation: 0 $\to$ 100.

In the second test case, the shortest possible string Timur started with is 11, and he performed the following operation: 11 $\to$ 0111.

In the third test case, the shortest possible string Timur started with is 10101, and he didn't perform any operations.

In the fourth test case, the shortest possible string Timur started with is the empty string (which we denote by ε), and he performed the following operations: $\varepsilon \to$ 10 $\to$ 0101 $\to$ 101010.

In the fifth test case, the shortest possible string Timur started with is 101, and he performed the following operations: 101 $\to$ 01011 $\to$ 1010110.

E. Two Strings Swaps

2 seconds, 256 megabytes

You are given two strings a and b consisting of lowercase English letters, both of length n. The characters of both strings have indices from 1 to n, inclusive.

You are allowed to do the following *changes*:

- Choose any index i ($1 \leq i \leq n$) and swap characters a_i and b_i;
- Choose any index i ($1 \leq i \leq n$) and swap characters a_i and a_{n-i+1};
- Choose any index i ($1 \leq i \leq n$) and swap characters b_i and b_{n-i+1}.

Note that if n is odd, you are formally allowed to swap $a_{\lceil \frac{n}{2} \rceil}$ with $a_{\lceil \frac{n}{2} \rceil}$ (and the same with the string b) but this move is useless. Also you can swap two equal characters but this operation is useless as well.

You have to make these strings equal by applying any number of *changes* described above, in any order. But it is obvious that it may be impossible to make two strings equal by these swaps.

In one *preprocess move* you can replace a character in a with another character. In other words, in a single *preprocess move* you can choose any index i ($1 \leq i \leq n$), any character c and set $a_i := c$.

Your task is to find the minimum number of *preprocess moves* to apply in such a way that after them you can make strings a and b equal by applying some number of *changes* described in the list above.

Note that the number of *changes* you make after the *preprocess moves* does not matter. Also note that you cannot apply *preprocess moves* to the string b or make any *preprocess moves* after the first *change* is made.

Input

The first line of the input contains one integer n ($1 \leq n \leq 10^5$) — the length of strings a and b.

The second line contains the string a consisting of exactly n lowercase English letters.

The third line contains the string b consisting of exactly n lowercase English letters.

Output

Print a single integer — the minimum number of *preprocess moves* to apply before *changes*, so that it is possible to make the string a equal to string b with a sequence of *changes* from the list above.

input

```
7
abacaba
bacabaa
```

output

```
4
```

input

```
5
zcabd
dbacz
```

output

```
0
```

In the first example *preprocess moves* are as follows: a_1 :='b', a_3 :='c', a_4 :='a' and a_5 :='b'. Afterwards, a ="bbcabba". Then we can obtain equal strings by the following sequence of *changes*: $swap(a_2, b_2)$ and $swap(a_2, a_6)$. There is no way to use fewer than 4 *preprocess moves* before a sequence of *changes* to make string equal, so the answer in this example is 4.

In the second example no *preprocess moves* are required. We can use the following sequence of *changes* to make a and b equal: $swap(b_1, b_5)$, $swap(a_2, a_4)$.

F. Han Solo and Lazer Gun

1 second, 256 megabytes

There are n Imperial stormtroopers on the field. The battle field is a plane with Cartesian coordinate system. Each stormtrooper is associated with his coordinates (x, y) on this plane.

Han Solo has the newest duplex lazer gun to fight these stormtroopers. It is situated at the point (x_0, y_0). In one shot it can can destroy all the stormtroopers, situated on some line that crosses point (x_0, y_0).

Your task is to determine what minimum number of shots Han Solo needs to defeat all the stormtroopers.

The gun is the newest invention, it shoots very quickly and even after a very large number of shots the stormtroopers don't have enough time to realize what's happening and change their location.

Input

The first line contains three integers n, x_0 и y_0 ($1 \le n \le 1000$, $-10^4 \le x_0, y_0 \le 10^4$) — the number of stormtroopers on the battle field and the coordinates of your gun.

Next n lines contain two integers each x_i, y_i ($-10^4 \le x_i, y_i \le 10^4$) — the coordinates of the stormtroopers on the battlefield. It is guaranteed that no stormtrooper stands at the same point with the gun. Multiple stormtroopers can stand at the same point.

Output

Print a single integer — the minimum number of shots Han Solo needs to destroy all the stormtroopers.

input

```
4 0 0
1 1
2 2
2 0
-1 -1
```

output

```
2
```

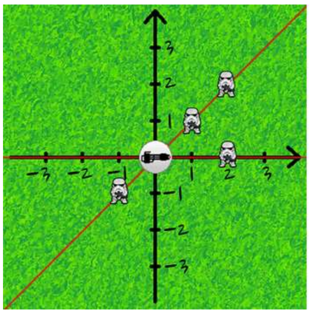
input

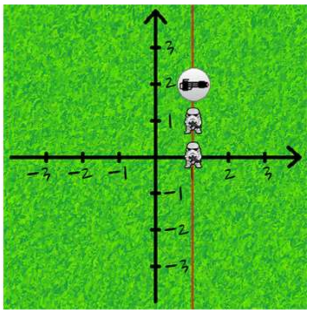
```
2 1 2
1 1
1 0
```

output

```
1
```

Explanation to the first and second samples from the statement, respectively:





G. Productive Meeting

2 seconds, 256 megabytes

An important meeting is to be held and there are exactly n people invited. At any moment, any two people can step back and talk in private. The same two people can talk several (as many as they want) times per meeting.

Each person has limited *sociability*. The sociability of the i-th person is a non-negative integer a_i. This means that after exactly a_i talks this person leaves the meeting (and does not talk to anyone else anymore). If $a_i = 0$, the i-th person leaves the meeting immediately after it starts.

A meeting is considered most *productive* if the maximum possible number of talks took place during it.

You are given an array of sociability a, determine which people should talk to each other so that the total number of talks is as large as possible.

Input

The first line contains an integer t ($1 \le t \le 1000$) — the number of test cases.

The next $2t$ lines contain descriptions of the test cases.

The first line of each test case description contains an integer n ($2 \le n \le 2 \cdot 10^5$) —the number of people in the meeting. The second line consists of n space-separated integers $a_1, a_2, \dots, a_n$ ($0 \le a_i \le 2 \cdot 10^5$) — the sociability parameters of all people.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$. It is also guaranteed that the sum of all a_i (over all test cases and all i) does not exceed $2 \cdot 10^5$.

Output

Print t answers to all test cases.

On the first line of each answer print the number k — the maximum number of talks possible in a meeting.

On each of the next k lines print two integers i and j ($1 \le i, j \le n$ and $i \ne j$) — the numbers of people who will have another talk.

If there are several possible answers, you may print any of them.

input

```
8
2
2 3
3
1 2 3
4
1 2 3 4
3
0 0 2
2
6 2
3
0 0 2
5
8 2 0 1 1
5
0 1 0 0 6
```

output

```
2
1 2
1 2
3
1 3
2 3
2 3
5
1 3
2 4
2 4
3 4
3 4
0
2
1 2
1 2
0
4
1 2
1 5
1 4
1 2
1
5 2
```

H. Elemental Decompress

2 seconds, 256 megabytes

You are given an array a of n integers.

Find two permutations[†] p and q of length n such that $\max(p_i, q_i) = a_i$ for all $1 \le i \le n$ or report that such p and q do not exist.

[†] A permutation of length n is an array consisting of n distinct integers from 1 to n in arbitrary order. For example, $[2, 3, 1, 5, 4]$ is a permutation, but $[1, 2, 2]$ is not a permutation (2 appears twice in the array), and $[1, 3, 4]$ is also not a permutation ($n = 3$ but there is 4 in the array).

Input

The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases. The description of test cases follows.

The first line of each test case contains a single integer n ($1 \le n \le 2 \cdot 10^5$).

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le n$) — the array a.

It is guaranteed that the total sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, if there do not exist p and q that satisfy the conditions, output "NO" (without quotes).

Otherwise, output "YES" (without quotes) and then output 2 lines. The first line should contain n integers $p_1, p_2, \ldots, p_n$ and the second line should contain n integers $q_1, q_2, \ldots, q_n$.

If there are multiple solutions, you may output any of them.

You can output "YES" and "NO" in any case (for example, strings "yEs", "yes" and "Yes" will be recognized as a positive response).

input

```
3
1
1
5
5 3 4 2 5
2
1 1
```

output

```
YES
1
1
YES
1 3 4 2 5
5 2 3 1 4
NO
```

In the first test case, $p = q = [1]$. It is correct since $a_1 = max(p_1, q_1) = 1$.

In the second test case, $p = [1, 3, 4, 2, 5]$ and $q = [5, 2, 3, 1, 4]$. It is correct since:

- $a_1 = \max(p_1, q_1) = \max(1, 5) = 5$,
- $a_2 = \max(p_2, q_2) = \max(3, 2) = 3$,
- $a_3 = \max(p_3, q_3) = \max(4, 3) = 4$,
- $a_4 = \max(p_4, q_4) = \max(2, 1) = 2$,
- $a_5 = \max(p_5, q_5) = \max(5, 4) = 5$.

In the third test case, one can show that no such p and q exist.

I. Koky's Birthday

2 seconds, 64 megabytes

It's $Koky's$ birthday and his mother is preparing some string as a gift for him. Right now she has a string S, she wants to make some changes on it to make it more beautiful. The problem is she is very busy and we don't know if she can finish the gift on time or not, so why don't you help her? Of course we don't want $Koky$ to be sad on his birthday!

$Koky's$ mother has written down the operations which will make the string more beautiful. There are three types of operations:

$1.\ X$, changing $Character_X$ to uppercase

$2.\ X$, changing $Character_X$ to lowercase

$3.\ X\ Y$, swapping $Character_X$ with $Character_Y$

Where X and Y are positions of characters in the initial string.

You will be given the initial string, and you have to execute the operations sequentially.

Input

First line of input consists of the initial String S, where S length won't exceed 10^4.

Second line contains only one number Q $(1 \le Q \le 10^4)$ $-$ Number of operations needed to be executed.

Next Q lines each contains an operation. If the operation type is 1 or 2 you will be given X, if type 3 you will be given X and Y.

Output

Print the final string - $Koky'sGift$ -

input

```
itskOkysbirthHdya
6
1 1
1 4
1 9
2 5
2 14
3 16 17
```

output

```
ItsKokysBirthhday
```

J. Order Book

2 seconds, 256 megabytes

In this task you need to process a set of stock exchange orders and use them to create *order book*.

An *order* is an instruction of some participant to buy or sell stocks on stock exchange. The order number i has price p_i, direction d_i — buy or sell, and integer q_i. This means that the participant is ready to buy or sell q_i stocks at price p_i for one stock. A value q_i is also known as a *volume* of an order.

All orders with the same price p and direction d are merged into one *aggregated* order with price p and direction d. The volume of such order is a sum of volumes of the initial orders.

An order book is a list of aggregated orders, the first part of which contains sell orders sorted by price in descending order, the second contains buy orders also sorted by price in descending order.

An order book of depth s contains s best aggregated orders for each direction. A buy order is better if it has higher price and a sell order is better if it has lower price. If there are less than s aggregated orders for some direction then all of them will be in the final order book.

You are given n stock exhange orders. Your task is to print order book of depth s for these orders.

Input

The input starts with two positive integers n and s $(1 \le n \le 1000, 1 \le s \le 50)$, the number of orders and the book depth.

Next n lines contains a letter d_i (either 'B' or 'S'), an integer p_i $(0 \le p_i \le 10^5)$ and an integer q_i $(1 \le q_i \le 10^4)$ — direction, price and volume respectively. The letter 'B' means buy, 'S' means sell. The price of any sell order is higher than the price of any buy order.

Output

Print no more than $2s$ lines with aggregated orders from order book of depth s. The output format for orders should be the same as in input.

input

```
6 2
B 10 3
S 50 2
S 40 1
S 50 6
B 20 4
B 25 10
```

output

```
S 50 8
S 40 1
B 25 10
B 20 4
```

Denote *(x, y)* an order with price x and volume y. There are 3 aggregated buy orders (10, 3), (20, 4), (25, 10) and two sell orders (50, 8), (40, 1) in the sample.

You need to print no more than two best orders for each direction, so you shouldn't print the order (10 3) having the worst price among buy orders.

K. Colourblindness

1 second, 256 megabytes

Vasya has a grid with 2 rows and n columns. He colours each cell red, green, or blue.

Vasya is colourblind and can't distinguish green from blue. Determine if Vasya will consider the two rows of the grid to be coloured the same.

Input

The input consists of multiple test cases. The first line contains an integer t $(1 \leq t \leq 100)$ — the number of test cases. The description of the test cases follows.

The first line of each test case contains an integer n $(1 \leq n \leq 100)$ — the number of columns of the grid.

The following two lines each contain a string consisting of n characters, each of which is either R, G, or B, representing a red, green, or blue cell, respectively — the description of the grid.

Output

For each test case, output "YES" if Vasya considers the grid's two rows to be identical, and "NO" otherwise.

You can output the answer in any case (for example, the strings "yEs", "yes", "Yes" and "YES" will be recognized as a positive answer).

input

```
6
2
RG
RB
4
GRBG
GBGB
5
GGGGG
BBBBB
7
BBBBBBB
RRRRRRR
8
RGBRRGBR
RGGRRBGR
1
G
G
```

output

```
YES
NO
YES
NO
YES
YES
```

In the first test case, Vasya sees the second cell of each row as the same because the second cell of the first row is green and the second cell of the second row is blue, so he can't distinguish these two cells. The rest of the rows are equal in colour. Therefore, Vasya will say that the two rows are coloured the same, even though they aren't.

In the second test case, Vasya can see that the two rows are different.

In the third test case, every cell is green or blue, so Vasya will think they are the same.

M. Chat Order

3 seconds, 256 megabytes

Polycarp is a big lover of killing time in social networks. A page with a chatlist in his favourite network is made so that when a message is sent to some friend, his friend's chat rises to the very top of the page. The relative order of the other chats doesn't change. If there was no chat with this friend before, then a new chat is simply inserted to the top of the list.

Assuming that the chat list is initially empty, given the sequence of Polycaprus' messages make a list of chats after all of his messages are processed. Assume that no friend wrote any message to Polycarpus.

Input

The first line contains integer n ($1 \le n \le 200\,000$) — the number of Polycarpus' messages. Next n lines enlist the message recipients in the order in which the messages were sent. The name of each participant is a non-empty sequence of lowercase English letters of length at most 10.

Output

Print all the recipients to who Polycarp talked to in the order of chats with them, from top to bottom.

input

```
4
alex
ivan
roman
ivan
```

output

```
ivan
roman
alex
```

input

```
8
alina
maria
ekaterina
darya
darya
ekaterina
maria
alina
```

output

```
alina
maria
ekaterina
darya
```

In the first test case Polycarpus first writes to friend by name "alex", and the list looks as follows:

1. alex

Then Polycarpus writes to friend by name "ivan" and the list looks as follows:

1. ivan
2. alex

Polycarpus writes the third message to friend by name "roman" and the list looks as follows:

1. roman
2. ivan
3. alex

Polycarpus writes the fourth message to friend by name "ivan", to who he has already sent a message, so the list of chats changes as follows:

1. ivan
2. roman
3. alex

N. Anya and Smartphone

1 second, 256 megabytes

Anya has bought a new smartphone that uses Berdroid operating system. The smartphone menu has exactly n applications, each application has its own icon. The icons are located on different screens, one screen contains k icons. The icons from the first to the k-th one are located on the first screen, from the $(k+1)$-th to the $2k$-th ones are on the second screen and so on (the last screen may be partially empty).

Initially the smartphone menu is showing the screen number 1. To launch the application with the icon located on the screen t, Anya needs to make the following gestures: first she scrolls to the required screen number t, by making t - 1 gestures (if the icon is on the screen t), and then make another gesture — press the icon of the required application exactly once to launch it.

After the application is launched, the menu returns to the first screen. That is, to launch the next application you need to scroll through the menu again starting from the screen number 1.

All applications are numbered from 1 to n. We know a certain order in which the icons of the applications are located in the menu at the beginning, but it changes as long as you use the operating system. Berdroid is intelligent system, so it changes the order of the icons by moving the more frequently used icons to the beginning of the list. Formally, right after an application is launched, Berdroid swaps the application icon and the icon of a preceding application (that is, the icon of an application on the position that is smaller by one in the order of menu). The preceding icon may possibly be located on the adjacent screen. The only exception is when the icon of the launched application already occupies the first place, in this case the icon arrangement doesn't change.

Anya has planned the order in which she will launch applications. How many gestures should Anya make to launch the applications in the planned order?

Note that one application may be launched multiple times.

Input

The first line of the input contains three numbers n, m, k $(1 \le n, m, k \le 10^5)$ — the number of applications that Anya has on her smartphone, the number of applications that will be launched and the number of icons that are located on the same screen.

The next line contains n integers, permutation $a_1, a_2, ..., a_n$ — the initial order of icons from left to right in the menu (from the first to the last one), a_i — is the id of the application, whose icon goes i-th in the menu. Each integer from 1 to n occurs exactly once among a_i.

The third line contains m integers $b_1, b_2, ..., b_m (1 \le b_i \le n)$ — the ids of the launched applications in the planned order. One application may be launched multiple times.

Output

Print a single number — the number of gestures that Anya needs to make to launch all the applications in the desired order.

input

```
8 3 3
1 2 3 4 5 6 7 8
7 8 1
```

output

```
7
```

input

```
5 4 2
3 1 5 2 4
4 4 4 4
```

output

```
8
```

In the first test the initial configuration looks like $(123)(456)(78)$, that is, the first screen contains icons of applications $1, 2, 3$, the second screen contains icons $4, 5, 6$, the third screen contains icons $7, 8$.

After application 7 is launched, we get the new arrangement of the icons — $(123)(457)(68)$. To launch it Anya makes 3 gestures.

After application 8 is launched, we get configuration $(123)(457)(86)$. To launch it Anya makes 3 gestures.

After application 1 is launched, the arrangement of icons in the menu doesn't change. To launch it Anya makes 1 gesture.

In total, Anya makes 7 gestures.

O. Balancing Brackets

2 seconds, 64 megabytes

A sequence of brackets is called balanced if one can turn it into a valid math expression by adding characters «+» and «1». For example, sequences «(())()», «()» and «(()(()))» are balanced, while «)(», «(()» and «(()))(» are not.

Given a sequence of brackets, find the minimum number of brackets that can be inserted into the sequence to make it balanced.

Input

The first line of the input contains an integer $T\,(1 \le T \le 100)$ – number of test cases.

The first line of each test case contains a single integer $N\,(1 \le N \le 10^4)$ – the length of the sequence.

The second line of each test case contains a string of length N, representing the sequence of brackets. Each character is either a '(' or a ')'.

Output

For each test case, print the minimum number of brackets that can be inserted into the sequence to make it balanced.

input

```
3
4
((()
5
))()(
6
(()())
```

output

```
2
3
0
```

P. Odd subarray

0.5 seconds, 128 megabytes

Given an array of size n, find the longest odd subarray of the array a.

Odd subarray is called odd if it has exactly one odd element.

A subarray of the array a is a sequence a_l, a_{l+1}, ..., a_r for some integers (l, r) such that $(1 \le l \le r \le n)$.

Input

The first line contains a single integer n $(1 \le n \le 2 \cdot 10^5)$, the size of the array.

The second line contains n integers $(1 \le a_i \le 10^9)$, elements of the array.

Output

If there is no answer print -1.

Otherwise print the answer in a single line, the length of the longest odd subarray.

input

```
6
1 5 4 2 3 2
```

output

```
4
```

input

```
4
8 12 6 10
```

output

```
-1
```

In the first example, the longest odd sub-array is $(3, 6)$.

In the second example, there are no odd numbers in the whole array so the answer is -1.

Q. Replace To Make Regular Bracket Sequence

1 second, 256 megabytes

You are given string s consists of opening and closing brackets of four kinds `<>`, `{}`, `[]`, `()`. There are two types of brackets: opening and closing. You can replace any bracket by another of the same type. For example, you can replace $<$ by the bracket `{`, but you can't replace it by `)` or $>$.

The following definition of a regular bracket sequence is well-known, so you can be familiar with it.

Let's define a regular bracket sequence (RBS). Empty string is RBS. Let s_1 and s_2 be a RBS then the strings $<s_1>s_2$, $\{s_1\}s_2$, $[s_1]s_2$, $(s_1)s_2$ are also RBS.

For example the string "`[[(){}]<>]`" is RBS, but the strings "`[)()`" and "`][()()`" are not.

Determine the least number of replaces to make the string s RBS.

Input

The only line contains a non empty string s, consisting of only opening and closing brackets of four kinds. The length of s does not exceed 10^6.

Output

If it's impossible to get RBS from s print `Impossible`.

Otherwise print the least number of replaces needed to get RBS from s.

input

```
[<}){}
```

output

```
2
```

input

```
{()}[]
```

output

```
0
```

input

```
]]
```

output

```
Impossible
```

R. String Manipulation 1.0

3 seconds, 256 megabytes

One popular website developed an unusual username editing procedure. One can change the username only by deleting some characters from it: to change the current name s, a user can pick number p and character c and delete the p-th occurrence of character c from the name. After the user changed his name, he can't undo the change.

For example, one can change name "`arca`" by removing the second occurrence of character "`a`" to get "`arc`".

Polycarpus learned that some user initially registered under nickname t, where t is a concatenation of k copies of string s. Also, Polycarpus knows the sequence of this user's name changes. Help Polycarpus figure out the user's final name.

Input

The first line contains an integer k $(1 \le k \le 2000)$. The second line contains a non-empty string s, consisting of lowercase Latin letters, at most 100 characters long. The third line contains an integer n $(0 \le n \le 20000)$ — the number of username changes. Each of the next n lines contains the actual changes, one per line. The changes are written as "p_i c_i" (without the quotes), where p_i $(1 \le p_i \le 200000)$ is the number of occurrences of letter c_i, c_i is a lowercase Latin letter. It is guaranteed that the operations are correct, that is, the letter to be deleted always exists, and after all operations not all letters are deleted from the name. The letters' occurrences are numbered starting from 1.

Output

Print a single string — the user's final name after all changes are applied to it.

input

```
2
bac
3
2 a
1 b
2 c
```

output

```
acb
```

input

```
1
abacaba
4
1 a
1 a
1 c
2 b
```

output

```
baa
```

Let's consider the first sample. Initially we have name "`bacbac`"; the first operation transforms it into "`bacbc`", the second one — to "`acbc`", and finally, the third one transforms it into "`acb`".

S. Little Girl and Maximum Sum

1 second, 256 megabytes

The little girl loves the problems on array queries very much.

One day she came across a rather well-known problem: you've got an array of n elements (the elements of the array are indexed starting from 1); also, there are q queries, each one is defined by a pair of integers l_i, r_i $(1 \le l_i \le r_i \le n)$. You need to find for each query the sum of elements of the array with indexes from l_i to r_i, inclusive.

The little girl found the problem rather boring. She decided to reorder the array elements before replying to the queries in a way that makes the sum of query replies maximum possible. Your task is to find the value of this maximum sum.

Input

The first line contains two space-separated integers n $(1 \le n \le 2 \cdot 10^5)$ and q $(1 \le q \le 2 \cdot 10^5)$ — the number of elements in the array and the number of queries, correspondingly.

The next line contains n space-separated integers a_i $(1 \le a_i \le 2 \cdot 10^5)$ — the array elements.

Each of the following q lines contains two space-separated integers l_i and r_i $(1 \le l_i \le r_i \le n)$ — the i-th query.

Output

In a single line print, a single integer — the maximum sum of query replies after the array elements are reordered.

Please, do not use the `%lld` specifier to read or write 64-bit integers in C++. It is preferred to use the `cin`, `cout` streams or the `%I64d` specifier.

input

```
3 3
5 3 2
1 2
2 3
1 3
```

output

```
25
```

input

```
5 3
5 2 4 1 3
1 5
2 3
2 3
```

output

```
33
```

T. Amir and Arithmophobia

2.5 seconds, 256 megabytes

$Isaac$ was very busy yesterday doing his tasks so he went to bed very late. He set his alarm at 8:00 AM, but he didn't wake up. Fortunately, his brother $Amir$ is energetic and wakes up early every day. He tried to wake $Isaac$ up to turn off the alarm, but, sadly he failed. $Amir$ was very annoyed by the alarm and decided to turn it off and let $Isaac$ sleep. When $Amir$ saw the alarm, he knew that he would fail. $Isaac$'s alarm can't be turned off unless some mathematical problems are solved. $Amir$ has $arithmophobia$ (fear of math and numbers). But, he hopes you don't have $arithmophobia$ too, and help him turn off the alarm.

Initially, $Amir$ has an empty array of numbers, the alarm gives you N commands, each command can be either:

1) $+\ x$ to add x to the array

2) $-\ x$ to remove x (if it exists) from the array

3) $--\ x$ to remove all numbers equal to x(if they exist) from the array

To close the alarm, you must print the average of numbers in the array after each command. Can you help $Amir$ turn off the alarm?

Input

In the first line you have N $(1 \le N \le 2 \cdot 10^5)$__ the number of commands.

The next N lines, each line has:

string s which can be either $(-, +, --)$ and X $(0 \le X \le 10^9)$ __ the commands as decribed above.

Output

Print N lines where each line represents the average of the numbers in the array, and if the array is empty, Print 0.

Your answer is considered correct if its absolute or relative error does not exceed 10^{-6}. Formally, let your answer be a, and the jury's answer be b.Your answer is accepted if and only if $\frac{|a-b|}{max(1,|b|)} \le 10^{-6}$.

input
10 + 60 -- 30 - 5 + 70 + 30 - 60 + 60 + 60 -- 60 + 20
output
60.000000 60.000000 60.000000 65.000000 53.333333 50.000000 53.333333 55.000000 50.000000 40.000000

The average is calculated by adding a group of numbers and then dividing by the count of those numbers

U. Letter

2 seconds, 256 megabytes

Vasya decided to write an anonymous letter cutting the letters out of a newspaper heading. He knows heading s_1 and text s_2 that he wants to send. Vasya can use every single heading letter no more than once. Vasya doesn't have to cut the spaces out of the heading — he just leaves some blank space to mark them. Help him; find out if he will manage to compose the needed text.

Input

The first line contains a newspaper heading s_1. The second line contains the letter text s_2. s_1 и s_2 are non-empty lines consisting of spaces, uppercase and lowercase Latin letters, whose lengths do not exceed 200 symbols. The uppercase and lowercase letters should be differentiated. Vasya does not cut spaces out of the heading.

Output

If Vasya can write the given anonymous letter, print YES, otherwise print NO

input
Instead of dogging Your footsteps it disappears but you dont notice anything where is your dog
output
NO

input
Instead of dogging Your footsteps it disappears but you dont notice anything Your dog is upstears
output
YES

input
Instead of dogging your footsteps it disappears but you dont notice anything Your dog is upstears
output
NO

input
abcdefg hijk k j i h g f e d c b a
output
YES

V. Glass Carving

2 seconds, 256 megabytes

Leonid wants to become a glass carver (the person who creates beautiful artworks by cutting the glass). He already has a rectangular w mm $\times$ h mm sheet of glass, a diamond glass cutter and lots of enthusiasm. What he lacks is understanding of what to carve and how.

In order not to waste time, he decided to practice the technique of carving. To do this, he makes vertical and horizontal cuts through the entire sheet. This process results in making smaller rectangular fragments of glass. Leonid does not move the newly made glass fragments. In particular, a cut divides each fragment of glass that it goes through into smaller fragments.

After each cut Leonid tries to determine what area the largest of the currently available glass fragments has. Since there appear more and more fragments, this question takes him more and more time and distracts him from the fascinating process.

Leonid offers to divide the labor — he will cut glass, and you will calculate the area of the maximum fragment after each cut. Do you agree?

Input

The first line contains three integers w, h, n $(2 \le w, h \le 200\ 000,$ $1 \le n \le 200\ 000)$.

Next n lines contain the descriptions of the cuts. Each description has the form $H\ y$ or $V\ x$. In the first case Leonid makes the horizontal cut at the distance y millimeters $(1 \le y \le h - 1)$ from the lower edge of the original sheet of glass. In the second case Leonid makes a vertical cut at distance x $(1 \le x \le w - 1)$ millimeters from the left edge of the original sheet of glass. It is guaranteed that Leonid won't make two identical cuts.

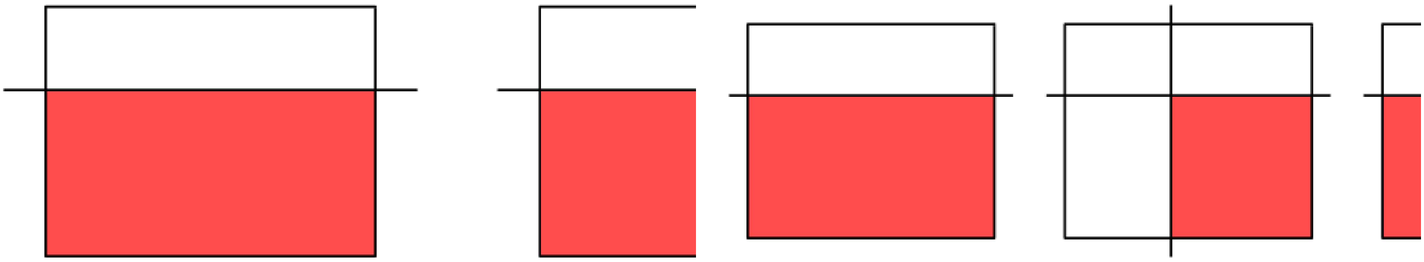
Output

After each cut print on a single line the area of the maximum available glass fragment in mm^2.

input
4 3 4 H 2 V 2 V 3 V 1
output
8 4 4 2

input
7 6 5 H 4 V 3 V 5 H 2 V 1
output
28 16 12 6 4

Picture for the first sample test:



Picture for the second sample test: